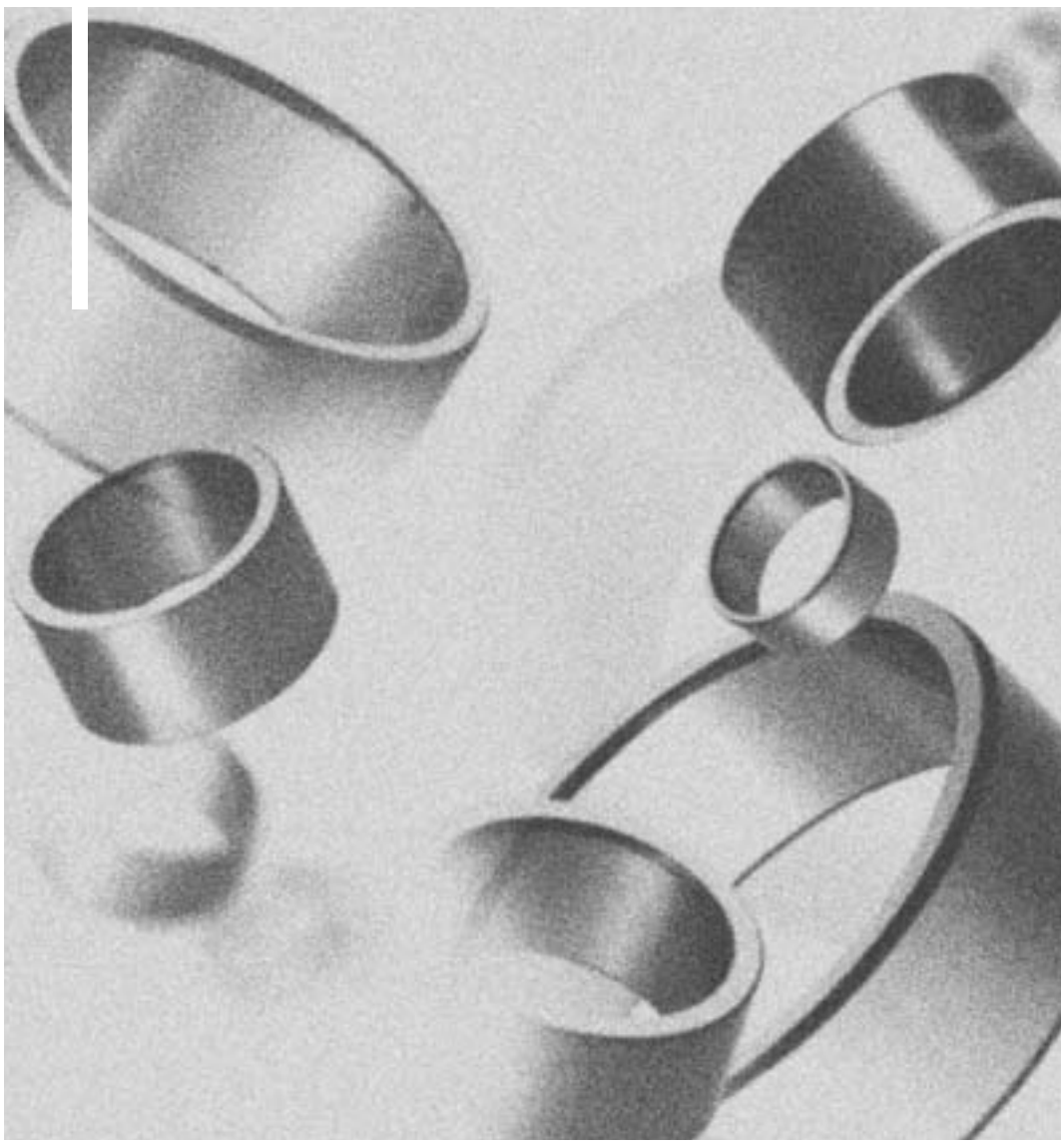
Inner Rings



Inner Rings

Inner Rings

Many of the needle roller bearings use a shaft as the direct raceway surface without using inner ring. However, it is recommended to use any of the inner rings described hereunder, together with needle roller bearing, where applicable shaft can not be surface-hardened and surface-finished by grinding. Any inner rings are made of high carbon chrome bearing steel and finished by grinding after heat-treated.

It is desirable to use an inner ring of wide width, where the axial displacement of a shaft is great and also a seal is used at the outer side of bearing.

Types and Designs

NTN inner rings are available in both of **IR type** with boundary dimensions of metric system and **MI type** with boundary dimensions of inch system. The both ends of inner ring raceway are chamfered in taper form to facilitate fitting of the inner ring. On the other hand, any inner rings with an asterisk ※-mark are fine-chamfered so as to allow greater axial displacement. In addition to these inner rings, manufacture of special inner ring type (with tail code: **D**) with oil hole on its center is also available.

Particularly where high running accuracy is required, on occasion the raceway surface of inner ring is finished by grinding after fitted on a shaft. In such a case, an inner ring with grinding allowance for its raceway can be supplied. Feel free to contact **NTN** for the detail of such an inner ring.

Composition of nominal number

Nominal number comprises type code (**IR** or **MI**), dimension code [bore dia. (d)×raceway dia. (F)×width (B)] and tail code.

The dimension of inch series **MI** type is expressed at the unit of 1/16 inch.

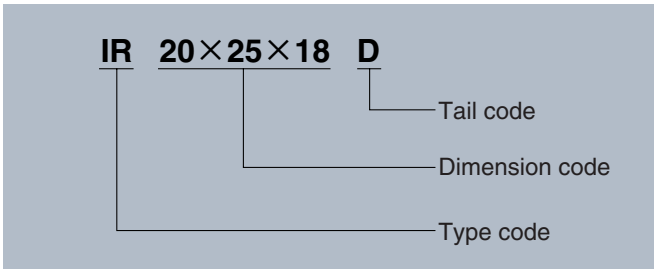


Fig. 1

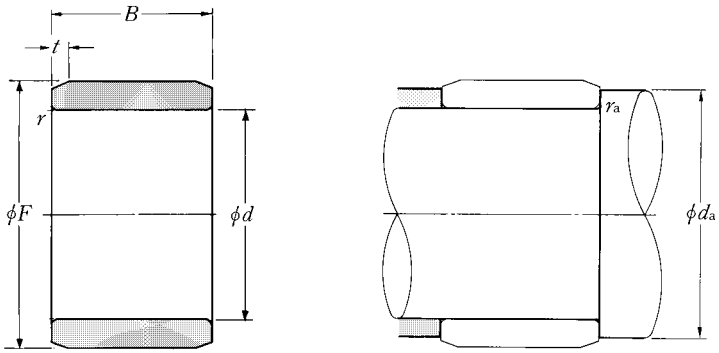
Dimensional accuracy for inner ring

The dimensional accuracy (bore diameter "d", width "B" and chamfer dimension "r_s min"), profile accuracy and running accuracy of the inner rings are as shown in **Tables 4.3** and 4.5 of Section 4 "**Bearing accuracy**" (page A-26). And the standard accuracy class of these items conforms to JIS Class-0, but other inner rings conforming to JIS Class-5 and -4 are also offerable on request.

For the dimensional tolerance for raceway diameter (F) shown in applicable Dimension Table, the inner ring is finished so the radial clearance comes to ordinary clearance (refer to **Table 5.1** of 5.2 "**Running clearance**" (page A-30) when it is combined with needle rollers.

Feel free to contact **NTN** for use of an inner ring subjected to radial clearance other than ordinary clearance.

Type IR

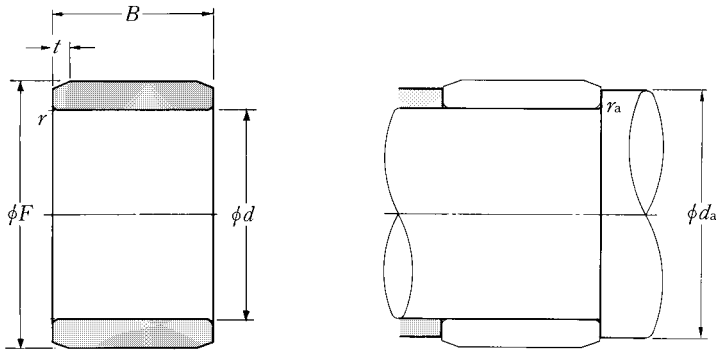


d 5～10mm

Boundary dimensions mm					Bearing numbers	Abutment dimensions mm		Mass kg
d	F [1]	B	$r_{s\,min}$ [2]	t		d_a min	r_{as} max	(approx.)
5	7	10	0.15	1	IR 5×7×10	6.2	0.15	0.0014
	8 $^{0}_{-0.006}$	12	0.3	1	IR 5×8×12	7	0.3	0.0028
	8	16	0.3	—	※IR 5×8×16	7	0.3	0.0038
6	8	10	0.15	1	IR 6×8×10	7.2	0.15	0.0017
	9	12	0.3	1	IR 6×9×12	8	0.3	0.0032
	9 $^{0}_{-0.006}$	16	0.3	1	IR 6×9×16	8	0.3	0.0043
	10	10	0.3	1	IR 6×10×10	8	0.3	0.0037
	10	12	0.3	—	※IR 6×10×12D	8	0.3	0.0046
	10	13	0.3	1	IR 6×10×13	8	0.3	0.0050
7	9	10	0.15	1	IR 7×9×10	8.2	0.15	0.0019
	10	10.5	0.3	1	IR 7×10×10.5	9	0.3	0.0031
	10 $^{0}_{-0.006}$	12	0.3	1	IR 7×10×12	9	0.3	0.0036
	10	16	0.3	1	IR 7×10×16	9	0.3	0.0049
	12 $^{-0}_{-0.008}$	16	0.3	1	IR 7×12×16	9	0.3	0.0093
8	10 $^{0}_{-0.006}$	11	0.15	1	IR 8×10×11	9.2	0.15	0.0024
	12	10	0.3	1	IR 8×12×10	10	0.3	0.0048
	12	10.5	0.3	1	IR 8×12×10.5	10	0.3	0.0050
	12 $^{0}_{-0.008}$	12	0.3	—	※IR 8×12×12D	10	0.3	0.0057
	12	12.5	0.3	1	IR 8×12×12.5	10	0.3	0.0059
	14	16	0.3	1	IR 8×14×16	10	0.3	0.013
9	12	11	0.3	1	IR 9×12×11	11	0.3	0.0041
	12 $^{0}_{-0.008}$	12	0.3	1	IR 9×12×12	11	0.3	0.0045
	12	16	0.3	1	IR 9×12×16	11	0.3	0.0061
	15	16	0.3	1	IR 9×15×16	11	0.3	0.014
10	13	12.5	0.3	1	IR10×13×12.5	12	0.3	0.0052
	14	12	0.3	1	IR10×14×12	12	0.3	0.0073
	14 $^{0}_{-0.008}$	13	0.3	1	IR10×14×13	12	0.3	0.0074
	14	14	0.3	—	※IR10×14×14D	12	0.3	0.0080
	14	16	0.3	—	※IR10×14×16	12	0.3	0.0092
	14	20	0.3	1	IR10×14×20	12	0.3	0.012

Note 1) Dimensional tolerance to secure ordinary clearance when any of these inner rings is combined with needle roller bearing with no inner ring.
2) Allowable minimum chamfer dimension r.
Remarks: 1. Nominal number plus code "**D**" represents inner ring with oil hole.
2. Nominal number plus ※-mark represents inner ring with fine-chamfered outer surface.

Type IR



d 10～15mm

Boundary dimensions mm					Bearing numbers	Abutment dimensions mm		Mass kg
d	F [1]	B	$r_{s\,min}$ [2]	t		d_a min	r_{as} max	(approx.)
	15	15.5	0.3	1	IR10×15×15.5	12	0.3	0.012
10	15 $^{0}_{-0.008}$	20.5	0.3	1	IR10×15×20.5	12	0.3	0.015
	16	16	0.3	1	IR10×16×16	12	0.3	0.015
	15	12	0.3	1	IR12×15×12	14	0.3	0.0058
	15	12.5	0.3	1	IR12×15×12.5	14	0.3	0.0061
	15	16	0.3	1	IR12×15×16	14	0.3	0.0078
	15	16.5	0.3	—	※IR12×15×16.5	14	0.3	0.0080
	15	22.5	0.3	—	※IR12×15×22.5	14	0.3	0.011
	16	12	0.3	1.5	IR12×16×12	14	0.3	0.0079
12	16	13	0.3	1.5	IR12×16×13	14	0.3	0.0087
	16 $^{0}_{-0.008}$	14	0.3	—	※IR12×16×14D	14	0.3	0.0095
	16	16	0.3	1.5	IR12×16×16	14	0.3	0.011
	16	20	0.3	1.5	IR12×16×20	14	0.3	0.014
	16	22	0.3	1.5	IR12×16×22	14	0.3	0.015
	17	20.5	0.3	1.5	IR12×17×20.5	14	0.3	0.019
	17	25.5	0.3	1.5	IR12×17×25.5	14	0.3	0.024
	18	16	0.3	1.5	IR12×18×16	14	0.3	0.018
14	17 $^{0}_{-0.008}$	17	0.3	1.5	IR14×17×17	16	0.3	0.0095
	18	12.5	0.3	1.5	IR15×18×12.5	17	0.3	0.0072
	18	16	0.3	1.5	IR15×18×16	17	0.3	0.0093
	18	16.5	0.3	1.5	IR15×18×16.5	17	0.3	0.0096
	18 $^{0}_{-0.008}$	17.5	0.3	1.5	IR15×18×17.5	17	0.3	0.010
	18	20.5	0.3	1.5	IR15×18×20.5	17	0.3	0.012
	18	25.5	0.3	1.5	IR15×18×25.5	17	0.3	0.015
15	19	16	0.3	1.5	IR15×19×16	17	0.3	0.013
	19	20	0.3	1.5	IR15×19×20	17	0.3	0.016
	20	12	0.3	1.5	IR15×20×12	17	0.3	0.012
	20 $^{0}_{-0.009}$	13	0.3	1.5	IR15×20×13	17	0.3	0.014
	20	14	0.3	—	※IR15×20×14D	17	0.3	0.015
	20	18	0.3	1.5	IR15×20×18	17	0.3	0.019
	20	20.5	0.3	1.5	IR15×20×20.5	17	0.3	0.021

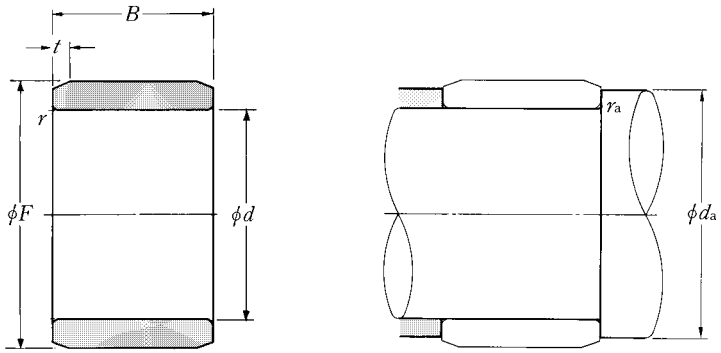
Note 1) Dimensional tolerance to secure ordinary clearance when any of these inner rings is combined with needle roller bearing with no inner ring.
2) Allowable minimum chamfer dimension r.
Remarks: 1. Nominal number plus code "**D**" represents inner ring with oil hole.
2. Nominal number plus ※-mark represents inner ring with fine-chamfered outer surface.

d 15～20mm

Boundary dimensions mm					Bearing numbers	Abutment dimensions mm		Mass kg
d	F [1]	B	$r_{s\,min}$ [2]	t		d_a min	r_{as} max	(approx.)
15	20	23	0.3	—	※IR15×20×23	17	0.3	0.024
	20 $^{0}_{-0.009}$	26	0.3	1.5	IR15×20×26	17	0.3	0.027
	20	30.5	0.3	1.5	IR15×20×30.5	17	0.3	0.032
	22	20	0.6	1.5	IR15×22×20	19	0.6	0.032
17	20	16	0.3	1.5	IR17×20×16	19	0.3	0.011
	20	16.5	0.3	1.5	IR17×20×16.5	19	0.3	0.011
	20	20	0.3	1.5	IR17×20×20	19	0.3	0.014
	20	20.5	0.3	—	※IR17×20×20.5	19	0.3	0.014
	20	30.5	0.3	—	※IR17×20×30.5	19	0.3	0.021
	21	16	0.3	1.5	IR17×21×16	19	0.3	0.014
	21	20	0.3	—	※IR17×21×20	19	0.3	0.018
	22 $^{0}_{-0.009}$	13	0.3	1.5	IR17×22×13	19	0.3	0.015
	22	14	0.3	—	※IR17×22×14D	19	0.3	0.016
	22	16	0.3	—	※IR17×22×16	19	0.3	0.019
	22	18	0.3	1.5	IR17×22×18	19	0.3	0.021
	22	20.5	0.3	1.5	IR17×22×20.5	19	0.3	0.024
	22	23	0.3	—	※IR17×22×23	19	0.3	0.027
	22	26	0.3	1.5	IR17×22×26	19	0.3	0.030
	22	32	0.3	1.5	IR17×22×32	19	0.3	0.036
	24	20	0.6	1.5	IR17×24×20	21	0.6	0.034
20	24	16	0.3	1.8	IR20×24×16	22	0.3	0.017
	24	20	0.3	—	※IR20×24×20	22	0.3	0.021
	24	28.5	0.3	—	※IR20×24×28.5	22	0.3	0.030
	25	12.5	0.3	1.8	IR20×25×12.5	22	0.3	0.016
	25	16	0.3	—	※IR20×25×16	22	0.3	0.021
	25 $^{0}_{-0.009}$	16.5	0.3	1.8	IR20×25×16.5	22	0.3	0.022
	25	17	0.3	1.8	IR20×25×17	22	0.3	0.022
	25	18	0.3	—	※IR20×25×18D	22	0.3	0.024
	25	20	0.3	—	※IR20×25×20	22	0.3	0.027
	25	20.5	0.3	1.8	IR20×25×20.5	22	0.3	0.028
	25	23	0.3	1.8	IR20×25×23	22	0.3	0.031

Note 1) Dimensional tolerance to secure ordinary clearance when any of these inner rings is combined with needle roller bearing with no inner ring.
2) Allowable minimum chamfer dimension r.
Remarks: 1. Nominal number plus code "**D**" represents inner ring with oil hole.
2. Nominal number plus ※-mark represents inner ring with fine-chamfered outer surface.

Type IR



d 20～28mm

Boundary dimensions mm					Bearing numbers	Abutment dimensions mm		Mass kg
d	F [1]	B	$r_{s\,min}$ [2]	t		d_a min	r_{as} max	(approx.)
	25	26	0.3	1.8	IR20×25×26	22	0.3	0.034
	25	26.5	0.3	—	※IR20×25×26.5	22	0.3	0.036
20	25 $^{0}_{-0.009}$	30	0.3	1.8	IR20×25×30	22	0.3	0.041
	25	32	0.3	1.8	IR20×25×32	22	0.3	0.041
	25	38.5	0.3	—	※IR20×25×38.5	22	0.3	0.053
	28	20	0.6	1.8	IR20×28×20	24	0.6	0.045
	26	16	0.3	1.8	IR22×26×16	24	0.3	0.017
	26	20	0.3	—	※IR22×26×20	24	0.3	0.022
	28	17	0.3	1.8	IR22×28×17	24	0.3	0.030
22	28 $^{0}_{-0.009}$	20	0.3	1.8	IR22×28×20	24	0.3	0.035
	28	20.5	0.3	1.8	IR22×28×20.5	24	0.3	0.036
	28	23	0.3	1.8	IR22×28×23	24	0.3	0.042
	28	30	0.3	—	※IR22×28×30	24	0.3	0.054
	29	20	0.3	—	※IR25×29×20	27	0.3	0.026
	29	30	0.3	1.8	IR25×29×30	27	0.3	0.039
	30	12.5	0.3	1.8	IR25×30×12.5	27	0.3	0.020
	30	16	0.3	1	IR25×30×16	27	0.3	0.024
	30	16.5	0.3	1.8	IR25×30×16.5	27	0.3	0.026
	30	17	0.3	1.8	IR25×30×17	27	0.3	0.027
	30	18	0.3	—	※IR25×30×18	27	0.3	0.030
	30 $^{0}_{-0.009}$	20	0.3	1.8	IR25×30×20	27	0.3	0.033
25	30	20.5	0.3	1.8	IR25×30×20.5	27	0.3	0.034
	30	23	0.3	1.8	IR25×30×23	27	0.3	0.038
	30	26	0.3	1.8	IR25×30×26	27	0.3	0.041
	30	26.5	0.3	—	※IR25×30×26.5	27	0.3	0.043
	30	30	0.3	1.8	IR25×30×30	27	0.3	0.050
	30	32	0.3	1	IR25×30×32	27	0.3	0.054
	30	38.5	0.3	—	※IR25×30×38.5	27	0.3	0.064
	32 $^{+0.008}_{-0.002}$	22	0.6	1.8	IR25×32×22	29	0.6	0.052
	32	30.5	0.6	1.8	IR25×32×30.5	29	0.6	0.072
28	32 $^{-0.008}_{-0.002}$	17	0.3	1.8	IR28×32×17	30	0.3	0.025

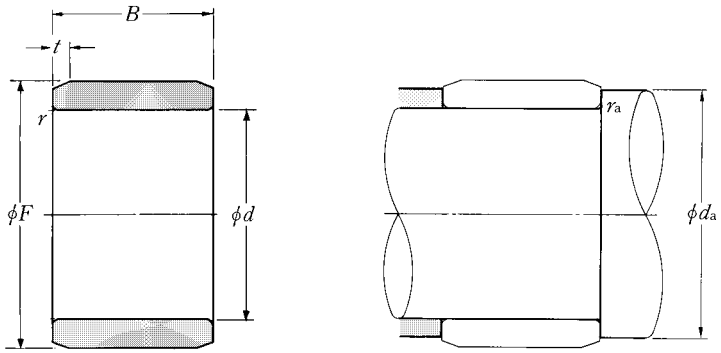
Note 1) Dimensional tolerance to secure ordinary clearance when any of these inner rings is combined with needle roller bearing with no inner ring.
2) Allowable minimum chamfer dimension r.
Remarks: 1. Nominal number plus ※-mark represents inner ring with fine-chamfered outer surface.

d 28～35mm

Boundary dimensions mm					Bearing numbers	Abutment dimensions mm		Mass kg
d	F [1]	B	$r_{s\,min}$ [2]	t		d_a min	r_{as} max	(approx.)
	32	20	0.3	1.8	IR28×32×20	30	0.3	0.028
28	32 $^{+0.008}_{-0.002}$	23	0.3	1.8	IR28×32×23	30	0.3	0.034
	32	30	0.3	—	※IR28×32×30	30	0.3	0.044
29	32 $^{+0.008}_{-0.002}$	13	0.3	1.8	IR29×32×13	31	0.3	0.015
	35	12.5	0.3	1.8	IR30×35×12.5	32	0.3	0.024
	35	13	0.3	1.3	IR30×35×13	32	0.3	0.025
	35	16	0.3	—	※IR30×35×16	32	0.3	0.031
	35	16.5	0.3	1.8	IR30×35×16.5	32	0.3	0.032
	35	17	0.3	1.8	IR30×35×17	32	0.3	0.032
	35	18	0.3	—	※IR30×35×16D	32	0.3	0.035
	35	20	0.3	—	※IR30×35×20	32	0.3	0.038
30	35 $^{+0.008}_{-0.002}$	20.5	0.3	1.8	IR30×35×20.5	32	0.3	0.039
	35	23	0.3	1.8	IR30×35×23	32	0.3	0.044
	35	26	0.3	1.8	IR30×35×26	32	0.3	0.050
	35	30	0.3	—	※IR30×35×30	32	0.3	0.059
	35	32	0.3	1.8	IR30×35×32	32	0.3	0.063
	37	18	0.3	1.8	IR30×37×18	32	0.3	0.050
	37	22	0.6	1.8	IR30×37×22	34	0.6	0.061
	38	20	0.6	—	※IR30×38×20	34	0.6	0.065
	37	20	0.3	2	IR32×37×20	34	0.3	0.040
	37	30	0.3	—	※IR32×37×30	34	0.3	0.063
32	38 $^{0}_{-0.011}$	32	0.3	2	IR32×38×32	34	0.3	0.082
	40	20	0.6	2	IR32×40×20	36	0.6	0.068
	40	27	0.6	2	IR32×40×27	36	0.6	0.092
	40	36	0.6	2	IR32×40×36	36	0.6	0.124
33	37 $^{0}_{-0.011}$	13	0.3	2	IR33×37×13	35	0.3	0.022
	40	12.5	0.3	2	IR35×40×12.5	37	0.3	0.027
	40	16.5	0.3	2	IR35×40×16.5	37	0.3	0.037
35	40 $^{0}_{-0.011}$	17	0.3	2	IR35×40×17	37	0.3	0.038
	40	20	0.3	2	IR35×40×20	37	0.3	0.044

Note 1) Dimensional tolerance to secure ordinary clearance when any of these inner rings is combined with needle roller bearing with no inner ring.
2) Allowable minimum chamfer dimension r.
Remarks: 1. Nominal number plus code "**D**" represents inner ring with oil hole.
2. Nominal number plus ※-mark represents inner ring with fine-chamfered outer surface.

Type IR



d 35～45mm

Boundary dimensions mm					Bearing numbers	Abutment dimensions mm		Mass kg
d	F [1]	B	$r_{s\,min}$ [2]	t		d_a min	r_{as} max	(approx.)
35	40	20.5	0.3	2	IR35×40×20.5	37	0.3	0.046
	40	30	0.3	—	※IR35×40×30	37	0.3	0.068
	40	34	0.3	1.8	IR35×40×34	37	0.3	0.079
	40	40	0.3	2	IR35×40×40	37	0.3	0.091
	42 $^{0}_{-0.011}$	20	0.6	1.8	IR35×42×20	39	0.6	0.064
	42	21	0.6	—	※IR35×42×21	39	0.6	0.068
	42	23	0.6	—	※IR35×42×23D	39	0.6	0.074
	42	27	0.6	2	IR35×42×27	39	0.6	0.080
	42	36	0.6	2	IR35×42×36	39	0.6	0.117
	43	22	0.6	2	IR35×43×22	39	0.6	0.081
38	43 $^{0}_{-0.011}$	20	0.3	1.8	IR38×43×20	40	0.3	0.048
	43	30	0.3	—	※IR38×43×30	40	0.3	0.074
40	45	16.5	0.3	2	IR40×45×16.5	42	0.3	0.042
	45	17	0.3	2	IR40×45×17	42	0.3	0.043
	45	20	0.3	2	IR40×45×20	42	0.3	0.051
	45	20.5	0.3	2	IR40×45×20.5	42	0.3	0.053
	45	26.5	0.3	—	※IR40×45×26.5	42	0.3	0.068
	45	30	0.3	2	IR40×45×30	42	0.3	0.077
	45 $^{0}_{-0.011}$	34	0.3	2	IR40×45×34	42	0.3	0.088
	45	40	0.3	2	IR40×45×40	42	0.3	0.106
	48	22	0.6	2	IR40×48×22	44	0.6	0.092
	48	23	0.6	—	※IR40×48×23	44	0.6	0.097
	48	30	0.6	2	IR40×48×30	44	0.6	0.123
	48	40	0.6	2	IR40×48×40	44	0.6	0.170
	50	20	0.3	0.8	IR40×50×20	44	0.3	0.106
	50	22	1	2	IR40×50×22	45	1	0.118
42	47 $^{0}_{-0.011}$	20	0.3	2	IR42×47×20	44	0.3	0.053
	47	30	0.3	2	IR42×47×30	44	0.3	0.080
45	50 $^{0}_{-0.011}$	20	0.3	2	IR45×50×20	47	0.3	0.057
	50	25	0.6	2	IR45×50×25	49	0.6	0.071

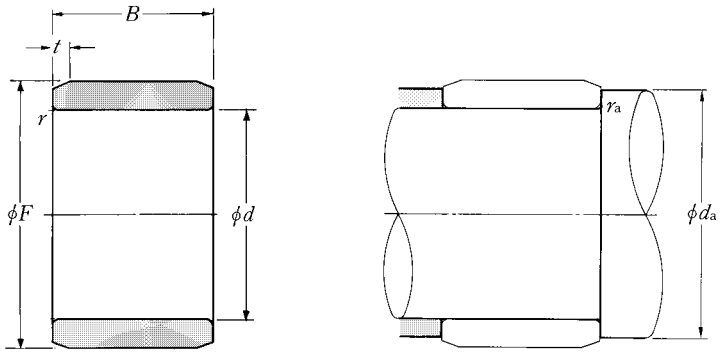
Note 1) Dimensional tolerance to secure ordinary clearance when any of these inner rings is combined with needle roller bearing with no inner ring.
2) Allowable minimum chamfer dimension r.
Remarks: 1. Nominal number plus code "**D**" represents inner ring with oil hole.
2. Nominal number plus ※-mark represents inner ring with fine-chamfered outer surface.

d 45～55mm

Boundary dimensions mm					Bearing numbers	Abutment dimensions mm		Mass kg
d	F [1]	B	$r_{s\,min}$ [2]	t		d_a min	r_{as} max	(approx.)
45	50	25.5	0.3	—	※IR45×50×25.5	47	0.3	0.074
	50 $^{0}_{-0.011}$	32	0.6	—	※IR45×50×32	49	0.6	0.092
	50	35	0.6	2	IR45×50×35	49	0.6	0.101
	50	40	0.3	1.5	IR45×50×40	47	0.3	0.115
	52	22	0.6	2	IR45×52×22	49	0.6	0.088
	52	23	0.6	—	※IR45×52×23D	49	0.6	0.093
	52	30	0.6	2	IR45×52×30	49	0.6	0.123
	52 $^{+0.008}_{-0.004}$	40	0.6	2	IR45×52×40	49	0.6	0.164
	55	20	0.6	2	IR45×55×20	49	0.6	0.116
	55	22	1	2	IR45×55×22	50	1	0.130
	55	40	0.6	2	IR45×55×40	49	0.6	0.173
50	55	20	0.6	2	IR50×55×20	54	0.6	0.063
	55	25	0.6	2	IR50×55×25	54	0.6	0.078
	55	35	0.6	2	IR50×55×35	54	0.6	0.112
	55	40	0.6	2	IR50×55×40	54	0.6	0.128
	58	22	0.6	2	IR50×58×22	54	0.6	0.113
	58 $^{+0.008}_{-0.004}$	23	0.6	—	※IR50×58×23D	54	0.6	0.119
	58	30	0.6	2	IR50×58×30	54	0.6	0.159
	58	40	0.6	2	IR50×58×40	54	0.6	0.209
	60	20	1	2	IR50×60×20	55	1	0.129
	60	25	1	2	IR50×60×25	55	1	0.163
	60	28	1.1	2	IR50×60×28	56.5	1	0.183
	60	40	1	2	IR50×60×40	55	1	0.262
55	60	25	0.6	2.2	IR55×60×25	59	0.6	0.086
	60	35	0.6	2	IR55×60×35	59	0.6	0.121
	63	25	1	2	IR55×63×25	60	1	0.141
	63 $^{0}_{-0.013}$	34	1	2.2	IR55×63×34	60	1	0.192
	63	45	1	2.2	IR55×63×45	60	1	0.256
	65	28	1.1	2.2	IR55×65×28	61.5	1	0.206
	65	30	1	2.2	IR55×65×30	60	1	0.220
	65	60	1	1.5	IR55×65×60	60	1	0.440

Note 1) Dimensional tolerance to secure ordinary clearance when any of these inner rings is combined with needle roller bearing with no inner ring.
2) Allowable minimum chamfer dimension r.
Remarks: 1. Nominal number plus code "**D**" represents inner ring with oil hole.
2. Nominal number plus ※-mark represents inner ring with fine-chamfered outer surface.

Type IR



d 60～75mm

Boundary dimensions mm					Bearing numbers	Abutment dimensions mm		Mass kg
d	F [1]	B	$r_{s\,min}$ [2]	t		d_a min	r_{as} max	(approx.)
60	68	25	1	2.2	IR60×68×25	65	1	0.152
	68	34	1	2.2	IR60×68×34	65	1	0.206
	68	35	0.6	2.2	IR60×68×35	64	0.6	0.213
	68 $^{0}_{-0.013}$	45	1	2.2	IR60×68×45	65	1	0.270
	70 $^{0}_{-0.013}$	25	1	2.2	IR60×70×25	65	1	0.195
	70	28	1.1	2.2	IR60×70×28	66.5	1	0.216
	70	30	1	2.2	IR60×70×30	65	1	0.232
	70	60	1	2.2	IR60×70×60	65	1	0.463
65	72	25	1	2.2	IR65×72×25	70	1	0.142
	72	34	1	2.2	IR65×72×34	70	1	0.193
	72	45	1	2.2	IR65×72×45	70	1	0.259
	73 $^{0}_{-0.013}$	25	0.6	2.2	IR65×73×25	69	0.6	0.164
	73 $^{0}_{-0.013}$	35	0.6	2.2	IR65×73×35	69	0.6	0.232
	75	28	1.1	2.2	IR65×75×28	71.5	1	0.240
	75	30	1	2.2	IR65×75×30	70	1	0.256
	75	60	1	2.2	IR65×75×60	70	1	0.513
70	80	25	1	2.2	IR70×80×25	75	1	0.224
	80	28	1.1	2.2	IR70×80×28	76.5	1	0.250
	80	30	1	2.2	IR70×80×30	75	1	0.267
	80 $^{0}_{-0.013}$	35	1	2.2	IR70×80×35	75	1	0.313
	80 $^{0}_{-0.013}$	40	1	2.2	IR70×80×40	75	1	0.358
	80	54	1	2.2	IR70×80×54	75	1	0.483
	80	56	1	2.2	IR70×80×56	75	1	0.502
	80	60	1	2.2	IR70×80×60	75	1	0.540
75	85	25	1	2.2	IR75×85×25	80	1	0.238
	85	30	1	2.2	IR75×85×30	80	1	0.287
	85 $^{0}_{-0.015}$	35	1	2.2	IR75×85×35	80	1	0.336
	85 $^{0}_{-0.015}$	40	1	2.2	IR75×85×40	80	1	0.385
	85	54	1	2.2	IR75×85×54	80	1	0.515
	90	32	1.1	2.2	IR75×90×32	81.5	1	0.480

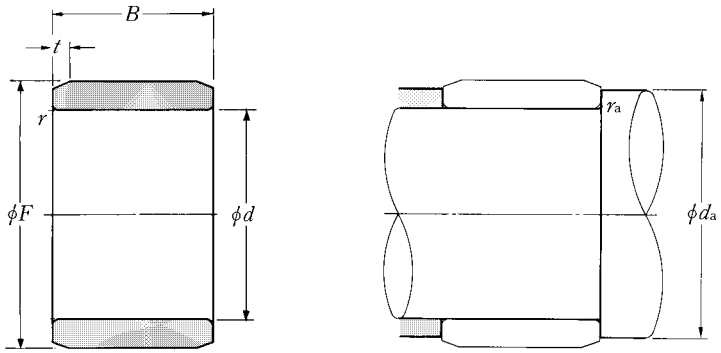
Note 1) Dimensional tolerance to secure ordinary clearance when any of these inner rings is combined with needle roller bearing with no inner ring.
2) Allowable minimum chamfer dimension r.

d 80～100mm

Boundary dimensions mm					Bearing numbers	Abutment dimensions mm		Mass kg
d	F [1]	B	$r_{s\,min}$ [2]	t		d_a min	r_{as} max	(approx.)
80	90	25	1	2.2	IR 80×90×25	85	1	0.254
	90	30	1	2.2	IR 80×90×30	85	1	0.304
	90 $^{0}_{-0.015}$	35	1	2.2	IR 80×90×35	85	1	0.355
	90	40	1	2.2	IR 80×90×40	85	1	0.408
	90	54	1	2.2	IR 80×90×54	85	1	0.543
	95	32	1.1	2.2	IR 80×95×32	86.5	1	0.510
85	95	26	1	2.5	IR 85×95×26	90	1	0.280
	95	30	1	2.5	IR 85×95×30	90	1	0.323
	95	36	1	2.5	IR 85×95×36	90	1	0.398
	100 $^{0}_{-0.015}$	32	1.1	2.5	IR 85×100×32	91.5	1	0.530
	100	35	1.1	2.5	IR 85×100×35	91.5	1	0.580
	100	46	1.1	2.5	IR 85×100×46	91.5	1	0.760
	100	63	1.1	2.5	IR 85×100×63	91.5	1	1.05
90	100	26	1	2.5	IR 90×100×26	95	1	0.294
	100	30	1	2.5	IR 90×100×30	95	1	0.340
	100	36	1	2.5	IR 90×100×36	95	1	0.406
	105 $^{0}_{-0.015}$	32	1.1	2.5	IR 90×105×32	96.5	1	0.560
	105	35	1.1	2.5	IR 90×105×35	96.5	1	0.610
	105	46	1.1	2.5	IR 90×105×46	96.5	1	0.800
	105	63	1.1	2.5	IR 90×105×63	96.5	1	1.11
95	105	26	1	2.5	IR 95×105×26	100	1	0.313
	105	36	1	2.5	IR 95×105×36	100	1	0.430
	110 $^{0}_{-0.015}$	32	1.1	2.5	IR 95×110×32	101.5	1	0.590
	110	35	1.1	2.5	IR 95×110×35	101.5	1	0.640
	110	46	1.1	2.5	IR 95×110×46	101.5	1	0.850
	110	63	1.1	2.5	IR 95×110×63	101.5	1	1.17
100	110	30	1.1	2.5	IR100×110×30	106.5	1	0.375
	110 $^{0}_{-0.015}$	40	1.1	2.5	IR100×110×40	106.5	1	0.505
	115	32	1.1	2.5	IR100×115×32	106.5	1	0.620
	115	40	1.1	2.5	IR100×115×40	106.5	1	0.775

Note 1) Dimensional tolerance to secure ordinary clearance when any of these inner rings is combined with needle roller bearing with no inner ring.
2) Allowable minimum chamfer dimension r.

Type IR



d 100～160mm

Boundary dimensions mm					Bearing numbers	Abutment dimensions mm		Mass kg
d	F [1]	B	$r_{s\,min}$ [2]	t		d_a min	r_{as} max	(approx.)
100	115 $^{0}_{-0.015}$	54	1.1	2.5	**IR100×115×54**	106.5	1	1.09
110	120 $^{0}_{-0.015}$	30	1	2.5	**IR110×120×30**	115	1	0.440
	120 $^{0}_{-0.015}$	40	1.1	2.5	**IR110×120×40**	116.5	1	0.580
	125 $^{0}_{-0.018}$	40	1.1	2.5	**IR110×125×40**	116.5	1	0.840
	125 $^{0}_{-0.018}$	54	1.1	2.5	**IR110×125×54**	116.5	1	1.16
120	130	30	1	2.2	**IR120×130×30**	125	1	0.440
	130	40	1.1	2.5	**IR120×130×40**	126.5	1	0.590
	135 $^{0}_{-0.018}$	40	2	2.5	**IR120×135×40**	129	2	0.870
	135	45	1.1	2.5	**IR120×135×45**	126.5	1	0.980
	135	60	1.1	2.5	**IR120×135×60**	126.5	1	1.25
130	145	32	1.5	3	**IR130×145×32**	138	1.5	0.780
	145	35	1.1	3	**IR130×145×35**	136.5	1	0.855
	145 $^{0}_{-0.018}$	42	1.5	3	**IR130×145×42**	138	1.5	1.05
	150 $^{0}_{-0.018}$	50	1.5	3	**IR130×150×50**	138	1.5	1.69
	150	52	2	3	**IR130×150×52**	139	2	1.75
	150	67	1.5	3	**IR130×150×67**	138	1.5	2.25
140	155	32	1.5	3	**IR140×155×32**	148	1.5	0.840
	155	35	1.1	3	**IR140×155×35**	146.5	1	0.917
	155 $^{0}_{-0.018}$	42	1.5	3	**IR140×155×42**	148	1.5	1.10
	160 $^{0}_{-0.018}$	50	1.5	3	**IR140×160×50**	148	1.5	1.70
	160	52	2	3	**IR140×160×52**	149	2	1.78
	160	67	1.5	3	**IR140×160×67**	148	1.5	2.30
150	165	32	1.5	3	**IR150×165×32**	158	1.5	0.900
	165	40	1.1	3	**IR150×165×40**	156.5	1	1.12
	165 $^{-0.017}_{-0.035}$	42	1.5	3	**IR150×165×42**	158	1.5	1.18
	170	52	2	3	**IR150×170×52**	159	2	2.00
	170	60	2	3	**IR150×170×60**	159	2	2.35
160	175 $^{-0.017}_{-0.035}$	40	1.1	3	**IR160×175×40**	166.5	1	1.20
	180 $^{-0.017}_{-0.035}$	60	2	3	**IR160×180×60**	169	2	2.50

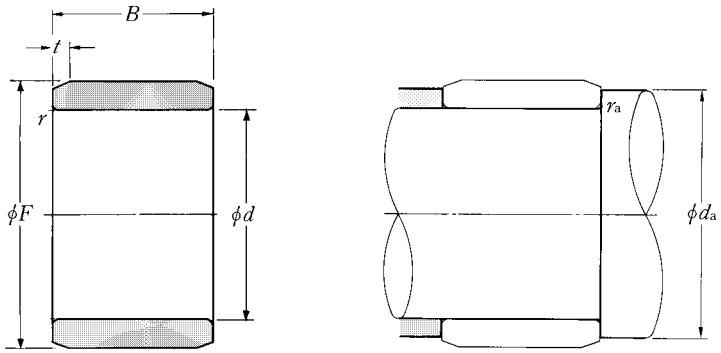
Note 1) Dimensional tolerance to secure ordinary clearance when any of these inner rings is combined with needle roller bearing with no inner ring.
2) Allowable minimum chamfer dimension r.

d 170～380mm

Boundary dimensions mm					Bearing numbers	Abutment dimensions mm		Mass kg
d	F [1]	B	$r_{s\,min}$ [2]	t		d_a min	r_{as} max	(approx.)
170	185 $^{-0.013}_{-0.043}$	45	1.1	3	IR170×185×45	176.5	1	1.45
	190 $^{-0.013}_{-0.043}$	60	2	3	IR170×190×60	179	2	2.65
180	195 $^{-0.013}_{-0.043}$	45	1.1	3	IR180×195×45	186.5	1	1.51
	205 $^{-0.013}_{-0.043}$	69	2	3	IR180×205×69	189	2	4.10
190	210 $^{-0.020}_{-0.050}$	50	1.5	3.5	IR190×210×50	198	1.5	2.41
	215 $^{-0.020}_{-0.050}$	69	2	3.5	IR190×215×69	199	2	4.10
200	220 $^{-0.020}_{-0.050}$	50	1.5	3.5	IR200×220×50	208	1.5	2.49
	225 $^{-0.020}_{-0.050}$	80	2.1	3.5	IR200×225×80	211	2	5.10
220	240 $^{-0.033}_{-0.063}$	50	1.5	3.5	IR220×240×50	228	1.5	2.75
	245 $^{-0.033}_{-0.063}$	80	2.1	3.5	IR220×245×80	231	2	5.70
240	265 $^{-0.037}_{-0.069}$	60	2	3.5	IR240×265×60	249	2	4.60
	265 $^{-0.037}_{-0.069}$	80	2.1	3.5	IR240×265×80	251	2	6.30
260	285 $^{-0.064}_{-0.092}$	60	2	4	IR260×285×60	269	2	4.98
	290 $^{-0.064}_{-0.092}$	100	2.1	4	IR260×290×100	271	2	10.0
280	305 $^{-0.064}_{-0.092}$	69	2	4	IR280×305×69	289	2	6.20
	310 $^{-0.064}_{-0.092}$	100	2.1	4	IR280×310×100	291	2	10.8
300	330 $^{-0.062}_{-0.098}$	80	2.1	4	IR300×330×80	311	2	9.30
	340 $^{-0.062}_{-0.098}$	118	3	4	IR300×340×118	313	2.5	18.5
320	350 $^{-0.062}_{-0.098}$	80	2.1	5	IR320×350×80	331	2	9.80
	360 $^{-0.062}_{-0.098}$	118	3	5	IR320×360×118	333	2.5	20.0
340	370 $^{-0.062}_{-0.098}$	80	2.1	5	IR340×370×80	351	2	10.1
	380 $^{-0.062}_{-0.098}$	118	3	5	IR340×380×118	353	2.5	22.0
360	390 $^{-0.090}_{-0.126}$	80	2.1	5	IR360×390×80	371	2	10.9
	400 $^{-0.090}_{-0.126}$	118	3	5	IR360×400×118	373	2.5	22.0
380	415 $^{-0.080}_{-0.120}$	100	2.1	5	IR380×415×100	391	2	18.5
	430 $^{-0.080}_{-0.120}$	140	4	5	IR380×430×140	396	3	35.0

Note 1) Dimensional tolerance to secure ordinary clearance when any of these inner rings is combined with needle roller bearing with no inner ring.
2) Allowable minimum chamfer dimension r.

Type IR



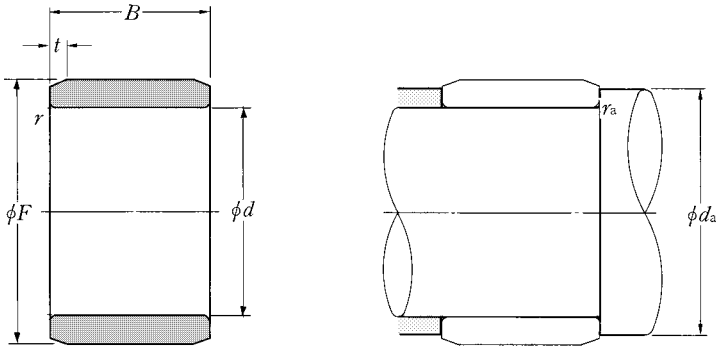
d 400～440mm

Boundary dimensions mm					Bearing numbers	Abutment dimensions mm		Mass kg
d	F [1]	B	$r_{s\,min}$ [2]	t		d_a min	r_{as} max	(approx.)
400	450 $^{-0.080}_{-0.120}$	140	4	5	**IR400×450×140**	416	3	36.5
420	470 $^{-0.105}_{-0.145}$	140	4	5	**IR420×470×140**	436	3	38.2
440	490 $^{-0.105}_{-0.145}$	160	4	5	**IR440×490×160**	456	3	46.5

Note 1) Dimensional tolerance to secure ordinary clearance when any of these inner rings is combined with needle roller bearing with no inner ring.
2) Allowable minimum chamfer dimension r.

Inch series

Type MI



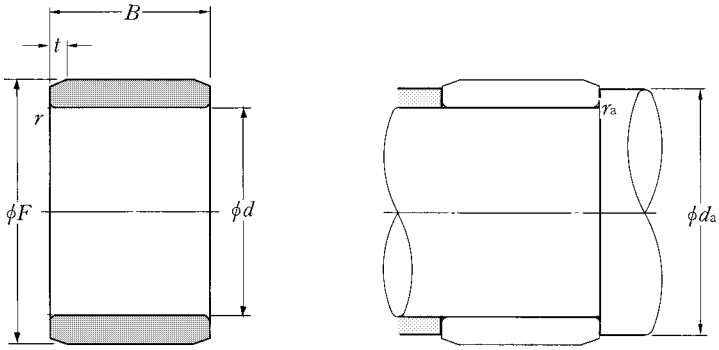
d z9.525～34.925mm

Boundary dimensions mm (1/25.4mm)					Bearing numbers	Abutment dimensions mm		Mass kg
d	*F* [1]	*B*	$r_{s\,min}$ [2]	*t*		d_a min	r_{as} max	(approx.)
9.525(3/8)	14.288(9/16)	12.95	0.6	—	※**MI-060908**	13.5	0.6	0.013
	15.875(5/8) 0 −0.008	19.30	0.6	1	**MI-061012**	13.5	0.6	0.019
	15.875(5/8)	25.65	0.6	1	**MI-061016**	13.5	0.6	0.025
12.700(1/2)	19.050(3/4)	16.13	1	—	※**MI-081210**	17.5	1	0.020
	19.050(3/4) 0 −0.009	19.30	1	—	※**MI-081212**	17.5	1	0.024
	19.050(3/4)	25.65	1	—	※**MI-081216**	17.5	1	0.032
15.875(5/8)	22.225(7/8)	9.78	1	—	※**MI-101406**	21	1	0.014
	22.225(7/8) 0 −0.009	12.95	1	—	※**MI-101408**	21	1	0.018
	22.225(7/8)	19.30	1	—	※**MI-101412**	21	1	0.027
	22.225(7/8)	25.65	1	—	※**MI-101416**	21	1	0.036
19.050(3/4)	25.400(1) 0 −0.009	19.30	1	—	※**MI-121612**	24	1	0.034
	25.400(1)	25.65	1	—	※**MI-121616**	24	1	0.045
22.225(7/8)	28.575(1 1/8)	12.95	1	—	※**MI-141808**	27	1	0.027
	28.575(1 1/8) 0 −0.009	19.30	1	—	※**MI-141812**	27	1	0.040
	28.575(1 1/8)	25.65	1	—	※**MI-141816**	27	1	0.052
	28.575(1 1/8)	32.00	1	—	※**MI-141820**	27	1	0.066
25.400(1)	31.750(1 1/4)	19.30	1	—	※**MI-162012**	30.5	1	0.039
	31.750(1 1/4) +0.008 −0.002	25.65	1	—	※**MI-162016**	30.5	1	0.052
	31.750(1 1/4)	32.00	1	—	※**MI-162020**	30.5	1	0.065
28.575(1 1/8)	34.925(1 3/8)	12.95	1	—	※**MI-182208**	33.5	1	0.032
	34.925(1 3/8) +0.008 −0.002	25.65	1	—	※**MI-182216**	33.5	1	0.063
	34.925(1 3/8)	32.00	1	—	※**MI-182220**	33.5	1	0.079
31.750(1 1/4)	38.100(1 1/2) 0 −0.011	25.65	1.5	—	※**MI-202416**	37	1.5	0.075
	38.100(1 1/2)	32.00	1.5	—	※**MI-202420**	37	1.5	0.094
33.338(1 5/16)	41.275(1 5/8) 0 −0.011	25.65	1.5	2	**MI-212616**	40	1.5	0.093
	41.275(1 5/8)	32.00	1.5	2	**MI-212620**	40	1.5	0.116
34.925(1 3/8)	41.275(1 5/8) 0 −0.011	16.13	1.5	2	**MI-222610**	40	1.5	0.073
	44.450(1 3/4)	25.65	1.5	2	**MI-222816**	41.5	1.5	0.117

Note 1) Dimensional tolerance to secure ordinary clearance when any of these inner rings is combined with needle roller bearing with no inner ring.
2) Allowable minimum chamfer dimension r.
Remarks: 1. Nominal number plus ※-mark represents inner ring with fine-chamfered outer surface. r

Inch series

Type MI



d 34.925～98.425mm

Boundary dimensions mm ($\frac{1}{25.4}$mm)					Bearing numbers	Abutment dimensions mm		Mass kg
d	F [1]	B	$r_{s\,min}$ [2]	t		d_a min	r_{as} max	(approx.)
34.925(1 3/8)	44.450(1 3/4) $^{0}_{-0.011}$	32.00	1.5	2	**MI-222820**	41.5	1.5	0.146
38.100(1 1/2)	44.450(1 3/4) $^{0}_{-0.011}$	19.30	1.5	2	**MI-242812**	43	1.5	0.062
	44.450(1 3/4) $^{0}_{-0.011}$	25.65	1.5	2	**MI-242816**	43	1.5	0.083
39.688(1 9/16)	47.625(1 7/8) $^{0}_{-0.011}$	32.00	1.5	2	**MI-253020**	46	1.5	0.136
41.275(1 5/8)	50.800(2) $^{+0.008}_{-0.004}$	25.65	1.5	2	**MI-263216**	48	1.5	0.140
	50.800(2) $^{+0.008}_{-0.004}$	32.00	1.5	2	**MI-263220**	48	1.5	0.175
44.450(1 3/4)	57.150(2 1/4) $^{+0.008}_{-0.004}$	38.35	1.5	2	**MI-283624**	51	1.5	0.310
	57.150(2 1/4) $^{+0.008}_{-0.004}$	44.70	1.5	2	**MI-283628**	51	1.5	0.360
50.800(2)	63.500(2 1/2) $^{0}_{-0.013}$	38.35	2	2	**MI-324024**	59	2	0.340
	63.500(2 1/2) $^{0}_{-0.013}$	44.70	2	2	**MI-324028**	59	2	0.420
57.150(2 1/4)	69.850(2 3/4)	25.65	2	2.2	**MI-364416**	65	2	0.257
	69.850(2 3/4) $^{0}_{-0.013}$	38.35	2	2.2	**MI-364424**	65	2	0.384
	69.850(2 3/4)	44.70	2	2.2	**MI-364428**	65	2	0.447
63.500(2 1/2)	76.200(3) $^{0}_{-0.013}$	38.35	2	2.2	**MI-404824**	71.5	2	0.417
	76.200(3) $^{0}_{-0.013}$	44.70	2	2.2	**MI-404828**	71.5	2	0.486
66.675(2 5/8)	82.550(3 1/4) $^{0}_{-0.015}$	44.70	2	2.2	**MI-425228**	74.5	2	0.648
	82.550(3 1/4) $^{0}_{-0.015}$	51.05	2	2.2	**MI-425232**	74.5	2	0.740
69.850(2 3/4)	82.550(3 1/4) $^{0}_{-0.015}$	44.70	2	2.2	**MI-445228**	78	2	0.530
76.200(3)	88.900(3 1/2) $^{0}_{-0.015}$	44.70	2	2.2	**MI-485628**	84	2	0.574
	88.900(3 1/2) $^{0}_{-0.015}$	51.05	2	2.2	**MI-485632**	84	2	0.655
79.375(3 1/8)	95.250(3 3/4) $^{0}_{-0.015}$	51.05	2.5	2.2	**MI-506032**	88.5	2.5	0.862
85.725(3 3/8)	101.600(4) $^{0}_{-0.015}$	51.05	2.5	2.5	**MI-546432**	94.5	2.5	0.930
92.075(3 5/8)	107.950(4 1/4) $^{0}_{-0.015}$	51.05	2.5	2.5	**MI-586832**	101	2.5	1.00
95.250(3 3/4)	114.300(4 1/2) $^{0}_{-0.015}$	57.40	2.5	2.5	**MI-607236**	104	2.5	1.40
98.425(3 7/8)	114.300(4 1/2) $^{0}_{-0.015}$	57.40	2.5	2.5	**MI-627236**	107.5	2.5	1.15
	114.300(4 1/2) $^{0}_{-0.015}$	63.88	2.5	2.5	**MI-627240**	107.5	2.5	1.28

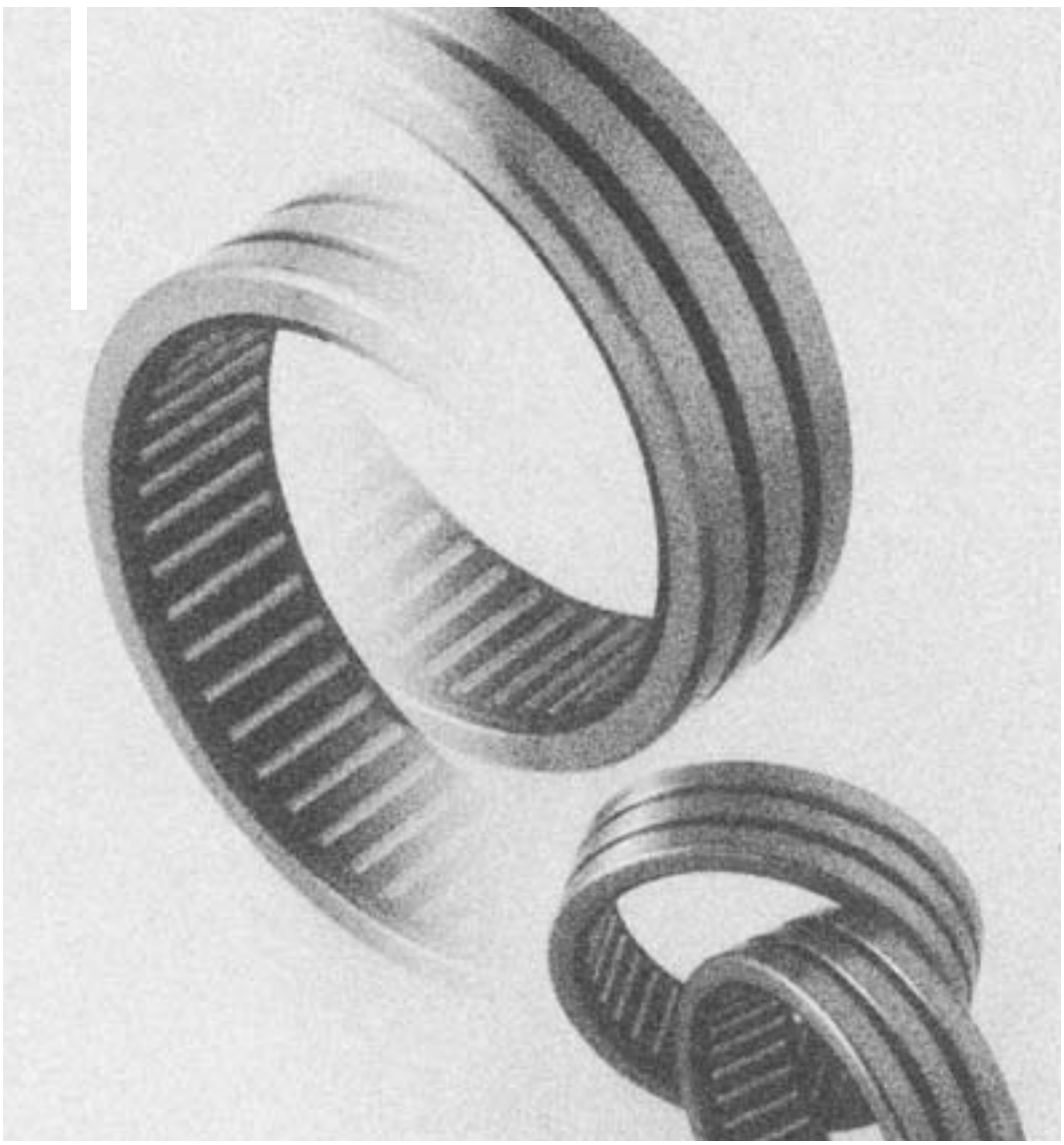
Note 1) Dimensional tolerance to secure ordinary clearance when any of these inner rings is combined with needle roller bearing with no inner ring.
2) Allowable minimum chamfer dimension r.

d 101.600～203.200mm

Boundary dimensions mm (1/25.4mm)					Bearing numbers	Abutment dimensions mm		Mass kg
d	F [1]	B	$r_{s\,min}$ [2]	t		d_a min	r_{as} max	(approx.)
	127.000(5)	51.05	2.5	2.5	**MI-648032**	110.5	2.5	1.82
101.600(4)	127.000(5) $^{0}_{-0.018}$	57.40	2.5	2.5	**MI-648036**	110.5	2.5	2.05
	127.000(5)	63.88	2.5	2.5	**MI-648040**	110.5	2.5	2.28
114.300(4 1/2)	139.700(5 1/2) $^{0}_{-0.018}$	63.88	2.5	2.5	**MI-728840**	123.5	2.5	2.58
	139.700(5 1/2)	76.58	2.5	2.5	**MI-728848**	123.5	2.5	3.10
120.650(4 3/4)	146.050(5 3/4) $^{0}_{-0.018}$	76.58	3	3	**MI-769248**	132	3	3.18
127.000(5)	152.400(6) $^{0}_{-0.018}$	63.88	3	3	**MI-809640**	140	3	2.80
	152.400(6)	76.58	3	3	**MI-809648**	140	3	3.35
139.700(5 1/2)	165.100(6 1/2) $^{-0.010}_{-0.028}$	63.88	3	3	**MI-8810440**	152.5	3	2.84
	165.100(6 1/2)	76.58	3	3	**MI-8810448**	152.5	3	3.40
152.400(6)	184.150(7 1/4) $^{-0.013}_{-0.033}$	76.58	3	3	**MI-9611648**	165.5	3	4.80
165.100(6 1/2)	196.850(7 3/4) $^{-0.013}_{-0.043}$	76.58	3	3	**MI-10412448**	178	3	5.30
177.800(7)	209.550(8 1/4) $^{-0.013}_{-0.043}$	76.58	3	3	**MI-11213248**	191	3	5.60
190.500(7 1/2)	222.250(8 3/4) $^{-0.020}_{-0.050}$	76.58	4	3	**MI-12014048**	206.5	4	6.10
203.200(8)	234.950(9 1/4) $^{-0.033}_{-0.063}$	76.58	4	3.5	**MI-12814848**	219	4	6.50

Note 1) Dimensional tolerance to secure ordinary clearance when any of these inner rings is combined with needle roller bearing with no inner ring.
2) Allowable minimum chamfer dimension r.

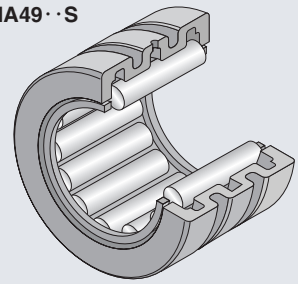
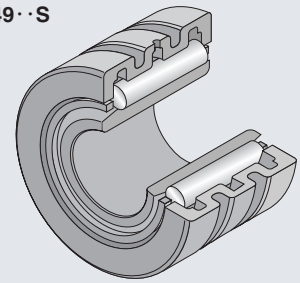
Adjustable-Clearance Needle Roller Bearings



Clearance-Adjustable Needle Roller Bearings

This bearing type comprises a solid outer ring with special machined section and needle rollers and cage built in the outer ring. The outer ring, needle rollers and cage are inseparable from each other. The raceway diameter of the outer ring is shrunk and the roller inscribed circle diameter (F_w) is reduced correspondingly by pressing the outer ring in axial direction.

The radial clearance is fine-adjustable by an load to be applied to the bearing in axial direction, that is, axial shrinkage of the outer ring. This bearing type is mainly applied to the work spindle of a machine tool etc. which require high running accuracy, whose accuracy class is JIS Class-4.

Bearing type	Applied shaft diameter (mm)	Composition of nominal bearing number	Remarks
Type RNA49··S 	ϕ30～ϕ125	**RNA 49 05 S** Type code Dimension series Bore diameter number Tail code **[Tail code]** S: Clearance-adjustable type	The dimensional accuracy, profile accuracy and running accuracy conform to JIS Class-4 so as to be available for applications requiring high running accuracy. Another inner ring type with grinding allowance for its raceway surface can also be supplied upon request.
Type NA49··S 	ϕ25～ϕ110		

Bearing accuracy

Application of this bearing type is limited to the portions of a machine which require high running accuracy. Therefore, the dimensional accuracy, profile accuracy and running accuracy of this bearing type conform to JIS Class-4. Higher running accuracy can be achieved by finishing the inner ring raceway surface with grinding allowance by grinding, after the inner ring was press-fitted on a shaft. Feel free to contact **NTN** for the more detailed information.

Radial clearance and bearing fits

The Type **NA49··S** with inner ring has a radial clearance smaller than the specific ordinary clearance, while the radial clearance of Type **RNA49··S** without inner ring is determined by shaft diameter being selected per applicable "Dimensions Table" which specifies the dimensional tolerances for the roller inscribed circle diameter (F_w).

The outer ring must be fitted in a housing by "close fit" or "tight fit" (interference of approximately 5 μm) to shrink the raceway diameter by pressing the outer ring in axial direction.

Bearing fit on a shaft and in a housing shall be per **Table 1.**

Table 1 Bearing fit in shaft and housing

Bearing type	Shaft	Housing
RNA49··S NA49··S	m5 k5	K4

Shaft and housing accuracy

This bearing type which is applied to the portions of a machine requiring high running accuracy is clamped with the outer ring inclination in installing restrained as less as possible. Therefore, the profile accuracy of the shaft and the housing must be made higher than that for general machined ring needle roller bearings. (**Table 2**)

Table 2 Shaft and housing accuracy

Characteristics	Shaft	Housing
Roundness (max)	IT2	IT3
Cylindricality (max)	IT2	IT3
Shoulder perpendicularity (max)	IT3	IT3

How to adjust radial clearance

The ratio of outer ring axial shrinkage to shrinkage of roller inscribed circle diameter (F_w) is nearly 3:1. In other words, the outer ring must be tightened by 3 mm in axial direction to make the radial clearance smaller by 1 mm.

For adjustment of the radial clearance, the outer ring must be tightened uniformly in parallel to the shaft center. In the case of adjustment by the method illustrated in **Fig.1**, a specific axial tightening value can be obtained by changing the thickness of a spacer, which was inserted between the housing and the lid, by lapping the spacer or replacing it with another spacer. In the case of adjustment by the method illustrated in **Fig.2**, the fixed ring 2 is fitted in the housing and, thereafter, the threaded lid 1 is screwed in until it comes in contact with the outer ring for zero setting. Then, the threaded lid 1 is further screwed in by the required value, after the fixed ring was loosened, and thereafter the fixed ring 2 is retightened. Further, it is recommended to use a saw-toothed tightening screw as illustrated in **Fig. 3** because accuracy-down can be minimized by use of such a screw.

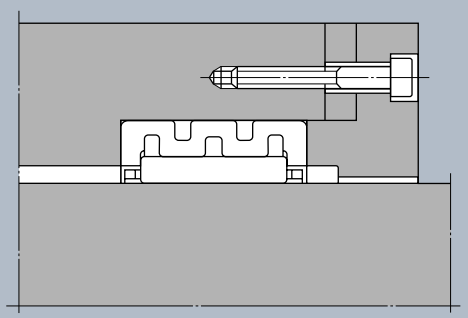


Fig. 1

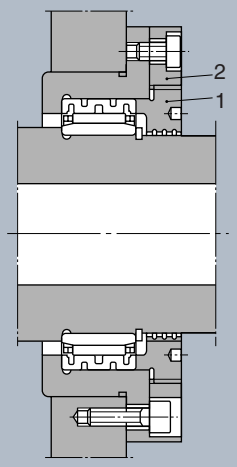


Fig. 2

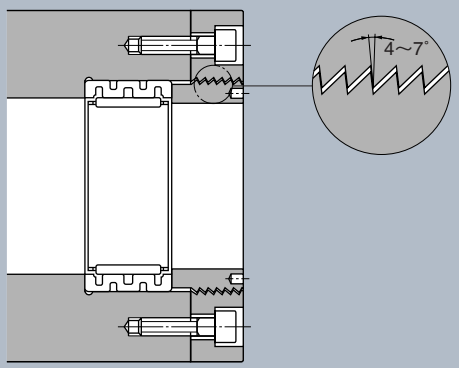
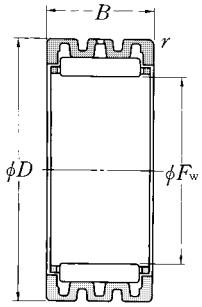
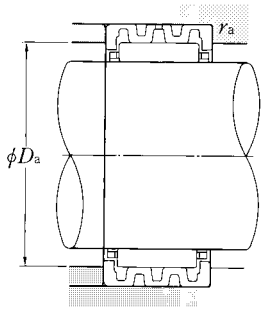


Fig. 3

Without inner ring

Type RNA49··S





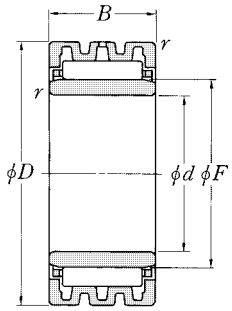
F_w 30～125mm

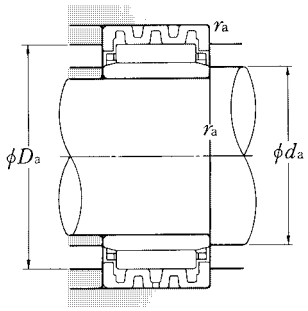
Boundary dimensions mm				Basic load ratings N		kgf		Limiting speeds r/min		Bearing numbers	Abutment dimensions mm		Mass kg
				dynamic	static	dynamic	static	grease	oil				
F_w	D	B	$r_{s\,min}$ 1)	C_r	C_{or}	C_r	C_{or}				D_a max	r_{as} max	(approx.)
30 $^{+0.031}_{+0.022}$	42	17	0.3	12 300	16 000	1 260	1 630	8 700	13 000	**RNA 4905S**	36	0.3	0.070
35 $^{+0.038}_{+0.028}$	47	17	0.3	14 900	21 600	1 520	2 200	7 300	11 000	**RNA 4906S**	41	0.3	0.086
40 $^{+0.038}_{+0.028}$	52	20	0.6	17 600	27 800	1 790	2 830	6 700	10 000	**RNA49/32S**	46	0.6	0.088
42 $^{+0.038}_{+0.028}$	55	20	0.6	18 300	29 800	1 870	3 050	6 300	9 500	**RNA 4907S**	48	0.6	0.099
48 $^{+0.038}_{+0.028}$	62	22	0.6	22 500	40 500	2 300	4 150	5 700	8 500	**RNA 4908S**	54	0.6	0.134
52 $^{+0.045}_{+0.035}$	68	22	0.6	23 100	43 500	2 360	4 400	5 000	7 500	**RNA 4909S**	58	0.6	0.168
58 $^{+0.045}_{+0.035}$	72	22	0.6	24 700	49 000	2 520	5 000	4 700	7 000	**RNA 4910S**	64	0.6	0.189
63 $^{+0.045}_{+0.035}$	80	25	1	33 000	65 500	3 350	6 650	4 300	6 500	**RNA 4911S**	70	1	0.212
68 $^{+0.045}_{+0.035}$	85	25	1	34 000	69 000	3 450	7 050	4 000	6 000	**RNA 4912S**	75	1	0.257
72 $^{+0.045}_{+0.035}$	90	25	1	34 000	70 500	3 450	7 200	3 700	5 500	**RNA 4913S**	79	1	0.286
80 $^{+0.045}_{+0.035}$	100	30	1	44 500	94 000	4 550	9 600	3 300	5 000	**RNA 4914S**	88	1	0.422
85 $^{+0.055}_{+0.040}$	105	30	1	45 500	99 000	4 650	10 100	3 100	4 700	**RNA 4915S**	93	1	0.451
90 $^{+0.055}_{+0.040}$	110	30	1	46 500	104 000	4 750	10 600	2 900	4 400	**RNA 4916S**	98	1	0.468
100 $^{+0.055}_{+0.040}$	120	35	1.1	57 000	140 000	5 800	14 300	2 700	4 000	**RNA 4917S**	108	1	0.594
105 $^{+0.055}_{+0.040}$	125	35	1.1	58 500	146 000	5 950	14 900	2 500	3 800	**RNA 4918S**	113	1	0.617
110 $^{+0.055}_{+0.040}$	130	35	1.1	59 500	152 000	6 050	15 500	2 400	3 600	**RNA 4919S**	118	1	0.735
115 $^{+0.055}_{+0.040}$	140	40	1.1	72 000	168 000	7 350	17 100	2 300	3 500	**RNA 4920S**	125	1	0.980
125 $^{+0.065}_{+0.050}$	150	40	1.1	75 500	184 000	7 700	18 800	2 100	3 200	**RNA 4922S**	135	1	1.04

Note 1) Allowable minimum chamfer dimension r.

With inner ring

Type NA49··S





d 25～110mm

Boundary dimensions mm					Basic load ratings dynamic N	static N	dynamic kgf	static kgf	Limiting speeds r/min grease	Limiting speeds r/min oil	Bearing numbers	Bearing clearanc before installation	Abutment dimensions mm			Mass kg
d	D	B	$r_{s\,min}$ [1]	F	C_r	C_{or}	C_r	C_{or}				μm	d_a min	D_a max	r_{as} max	(approx.)
25	42	17	0.3	30	12 300	16 000	1 260	1 630	8 700	13 000	**NA 4905S**	10～25	27	36	0.3	0.084
30	47	17	0.3	35	14 900	21 600	1 520	2 200	7 300	11 000	**NA 4906S**	12～25	32	41	0.3	0.099
32	52	20	0.6	40	17 600	27 800	1 790	2 830	6 700	10 000	**NA49/32S**	12～25	36	46	0.6	0.157
35	55	20	0.6	42	18 300	29 800	1 870	3 050	6 300	9 500	**NA 4907S**	15～30	39	48	0.6	0.164
40	62	22	0.6	48	22 500	40 500	2 300	4 150	5 700	8 500	**NA 4908S**	15～30	44	54	0.6	0.227
45	68	22	0.6	52	23 100	43 500	2 360	4 400	5 000	7 500	**NA 4909S**	15～35	49	58	0.6	0.257
50	72	22	0.6	58	24 700	49 000	2 520	5 000	4 700	7 000	**NA 4910S**	15～35	54	64	0.6	0.271
55	80	25	1	63	33 000	65 500	3 350	6 650	4 300	6 500	**NA 4911S**	15～35	60	70	1	0.382
60	85	25	1	68	34 000	69 000	3 450	7 050	4 000	6 000	**NA 4912S**	20～40	65	75	1	0.410
65	90	25	1	72	34 000	70 500	3 450	7 200	3 700	5 500	**NA 4913S**	20～40	70	79	1	0.427
70	100	30	1	80	44 500	94 000	4 550	9 600	3 300	5 000	**NA 4914S**	20～40	75	88	1	0.689
75	105	30	1	85	45 500	99 000	4 650	10 100	3 100	4 700	**NA 4915S**	25～45	80	93	1	0.740
80	110	30	1	90	46 500	104 000	4 750	10 600	2 900	4 400	**NA 4916S**	25～45	85	98	1	0.774
85	120	35	1.1	100	57 000	140 000	5 800	14 300	2 700	4 000	**NA 4917S**	25～45	91.5	108	1	1.18
90	125	35	1.1	105	58 500	146 000	5 950	14 900	2 500	3 800	**NA 4918S**	25～50	96.5	113	1	1.23
95	130	35	1.1	110	59 500	152 000	6 050	15 500	2 400	3 600	**NA 4919S**	25～50	101.5	118	1	1.40
100	140	40	1.1	115	72 000	168 000	7 350	17 100	2 300	3 500	**NA 4920S**	25～50	106.5	125	1	1.91
110	150	40	1.1	125	75 500	184 000	7 700	18 800	2 100	3 200	**NA 4922S**	30～60	116.5	135	1	2.12

Note 1) Allowable minimum chamfer dimension r.